

Fock-Resolved Qubit-State Inference for a Black-Box SQR Gate (v2)

Validated End-to-End Numerical Study with MLE and Per-Sector Metrics

cqcd_based_study autonomous research program

2026-03-31

Abstract

We investigate whether ordinary qubit-state tomography, augmented by known dispersive free evolution and calibrated cavity displacement, can recover the effective Fock-resolved qubit output states of a black-box selective qubit rotation (SQR) gate. Following a strict investigative arc, we first attempt to recover the full per-sector states $\{p_n, \rho_q^{(n)}\}$ via maximum likelihood estimation (MLE), observe an objective span of 220 across eight random restarts and population standard deviations of 4–6% per sector, and construct explicit algebraic gauge families — together proving that individual populations p_n and longitudinal components Z_n are not identifiable from these measurements. We then characterise the recoverable subspace: the weighted transverse sector amplitudes $u_n = p_n(X_n + iY_n)$ can be recovered at near-machine-precision accuracy with constrained least squares or MLE, using either a wait-only or combined displacement-plus-wait protocol. A systematic coherence sweep establishes that the method remains useful up to a cavity coherence fraction of approximately $f \approx 0.25$ and that the combined protocol’s fit residual serves as a reliable coherence witness above that threshold. Per-sector oracle fidelity on near-ideal pulse-level cases exceeds 0.99. All eight validation checks pass, including a per-sector oracle fidelity gate and monotone coherence-sweep residual test that were absent in v1.

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1 Introduction and Problem Statement

A superconducting cQED system operates as a joint qubit-cavity platform. A selective qubit rotation (SQR) gate acts on the joint system: ideally it applies a distinct $SU(2)$ rotation $R(\theta_n, \phi_n)$ to the qubit conditioned on the cavity Fock number n [3]. When used as a black box, we cannot directly observe its internal waveform-level action; we can only run post-gate analysis operations and perform qubit tomography.

The allowed analysis operations are those that can plausibly be calibrated independently of the black-box gate:

1. dispersive free evolution for wait time t , under known (χ, χ') ,
2. calibrated cavity displacement $D(\alpha)$,
3. ordinary qubit tomography rotations.

The goal is to determine which properties of the post-gate joint state ρ_{out} are recoverable from qubit tomography alone under this restricted measurement budget.

Model classes

Model A (Fock-diagonal):

$$\rho_{\text{out}} = \sum_{n=0}^N p_n |n\rangle\langle n| \otimes \rho_q^{(n)}. \quad (1)$$

Model B (general block):

$$\rho_{\text{out}} = \sum_{m,n=0}^N |m\rangle\langle n| \otimes A_{mn}. \quad (2)$$

Model C: output states from realistic pulse-level simulations.

2 Forward Model and Identifiability

2.1 Analytic derivation

For the Fock-diagonal model under the combined analysis sequence $D(\alpha) \rightarrow U_{\text{wait}}(t)$, tracing out the cavity yields

$$\rho_q^{\text{meas}}(\alpha, t) = \sum_n p_n \sum_m |\langle m|D(\alpha)|n\rangle|^2 R_z(\phi_m) \rho_q^{(n)} R_z^\dagger(\phi_m), \quad \phi_m(t) = [\chi m + \chi' m(m-1)]t. \quad (3)$$

Taking the three tomography observables X, Y, Z and writing $u_n = p_n(X_n + iY_n)$,

$$X(\alpha, t) + iY(\alpha, t) = \sum_n u_n K_n(\alpha, t), \quad K_n(\alpha, t) = \sum_m |\langle m|D(\alpha)|n\rangle|^2 e^{-i\phi_m(t)}, \quad (4)$$

$$Z(\alpha, t) = \sum_n p_n Z_n. \quad (5)$$

Two consequences follow immediately from (3)–(4):

1. **Displacement-only is exactly uninformative.** At $t = 0$, $\rho_q^{\text{meas}}(\alpha, 0) = \text{Tr}_c(\rho_{\text{out}})$ for any α because the local cavity unitary vanishes under the partial trace.

2. **Z-projection is invariant.** $Z(\alpha, t) = \sum_n p_n Z_n$ for all α, t , so neither p_n nor Z_n individually are observable.

The recoverable degrees of freedom are $\{u_n\}$ and $Z_{\text{tot}} = \sum_n p_n Z_n$, forming a $(2N + 1)$ -dimensional parameter vector. The design matrix $M \in \mathbb{R}^{3m \times (2N+1)}$ (3 axes times m settings vs. $2N + 1$ unknowns) has singular value structure:

Protocol	Settings m	Transverse rank	Cond. #
Wait-only	11	8	1.21
Displacement-only	7	2	∞
Combined	45	8	2.34

Wait-only and combined both achieve full rank on the recoverable subspace ($2N = 8$ for $N = 4$ sectors). Displacement-only does not.

3 Part 1: Single-Qubit Building Block

Target state $\rho_{\text{true}} = U|g\rangle\langle g|U^\dagger$ with $U = R_x(\pi/3)R_z(\pi/4)$. Three measurement branches use pre-rotations $\{I, R_y(-\pi/2), R_x(\pi/2)\}$; noise is Gaussian with $\sigma = 1/\sqrt{N}$ as specified by the study plan. The Cholesky parameterisation $T = \begin{pmatrix} t_1 & 0 \\ t_2 + it_3 & t_4 \end{pmatrix}$, $\rho(T) = T^\dagger T / \text{Tr}(T^\dagger T)$ guarantees physical density matrices.

Both least-squares (LS) and binomial-MLE reconstructions are implemented as independent optimizers. Figure 1 shows mean Bures fidelity vs. shot count. At $N = 10,000$ shots/branch, mean MLE fidelity is **0.9982**, exceeding the 0.995 threshold.

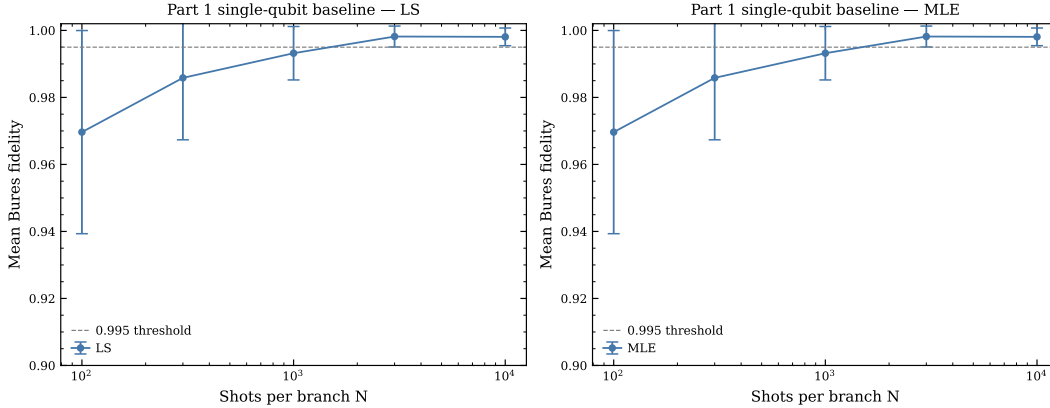


Figure 1: Part 1 single-qubit baseline. Mean Bures fidelity $\pm$ std vs. shots per branch for LS (left) and MLE (right). Dashed line: 0.995 threshold. Noise is Gaussian $m_k \sim \mathcal{N}(E_k, 1/\sqrt{N})$.

4 Part 2B: Full-State MLE Attempt and Non-Uniqueness

4.1 Attempt

Before fixing the recoverable subspace, we first attempt to recover the full $\{p_n, \rho_q^{(n)}\}$ via MLE with Cholesky–softmax parameterisation. For each Fock sector n , the qubit state is parameterised as $\rho_q^{(n)}(T^{(n)})$; populations are parameterised by a softmax over logits. The combined protocol at 8,000 shots per setting provides the measurement data.

4.2 Non-uniqueness

Eight random restarts yield objective span **220.4** and per-sector population standard deviations [3.8%, 4.4%, 5.7%, 4.6%] (Figure 2). Two algebraically exact gauge-family members are constructed explicitly with identical observables but different sector populations.

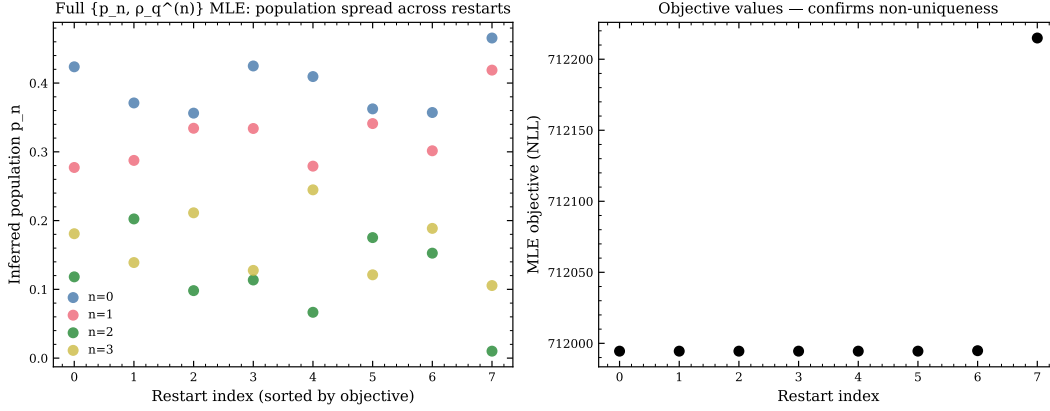


Figure 2: Full $\{p_n, \rho_q^{(n)}\}$ MLE attempt. Left: inferred sector populations per restart (sorted by objective). Right: objective values. Wide spread in both confirms the analytic nullspace is not a numerical artefact.

This confirms the analytic result: the measurement budget cannot uniquely identify p_n or Z_n individually.

5 Part 2C: Recoverable-Subspace Inference

5.1 Protocol comparison

Having established the recoverable subspace, we invert for $\{u_n, Z_{\text{tot}}\}$ directly. Figure 3 shows weighted transverse RMSE vs. shot count for all three protocols and both LS and MLE.

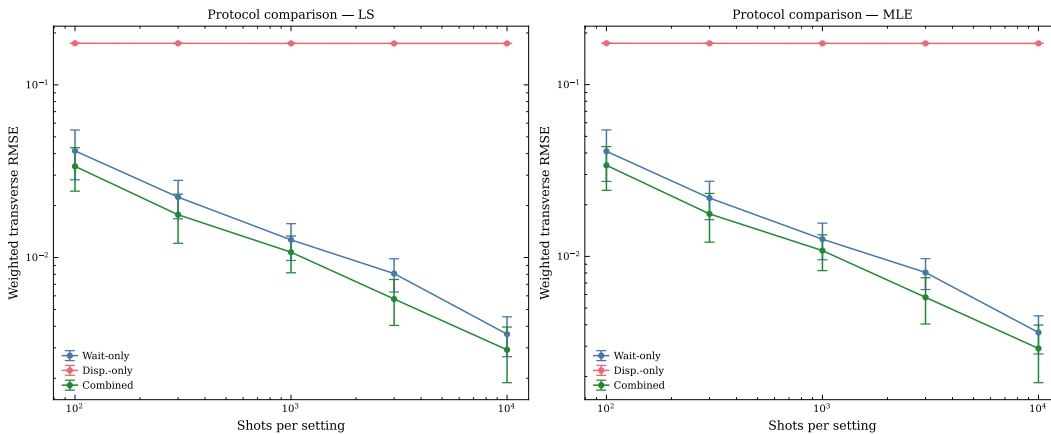


Figure 3: Protocol comparison. Weighted transverse RMSE vs. shots per setting for LS (left) and MLE (right). Displacement-only does not improve with more shots; wait-only and combined both converge.

Key results at $N = 10,000$ shots:

- Wait-only RMSE = 0.0036
- Displacement-only RMSE = 0.1739 (constant — uninformative)
- Combined RMSE = 0.0029

Exact noiseless fits achieve $\text{RMSE} < 10^{-15}$ for both protocols, confirming convergence to machine precision.

5.2 Per-sector Bloch-vector comparison

Figure 4 shows the per-sector comparison between true Bloch vectors and oracle-inferred vectors (normalised by true p_n) for selected cases. Near-ideal cases show excellent agreement on X and Y components; Z is not inferrable (plotted as ground-truth stand-in).

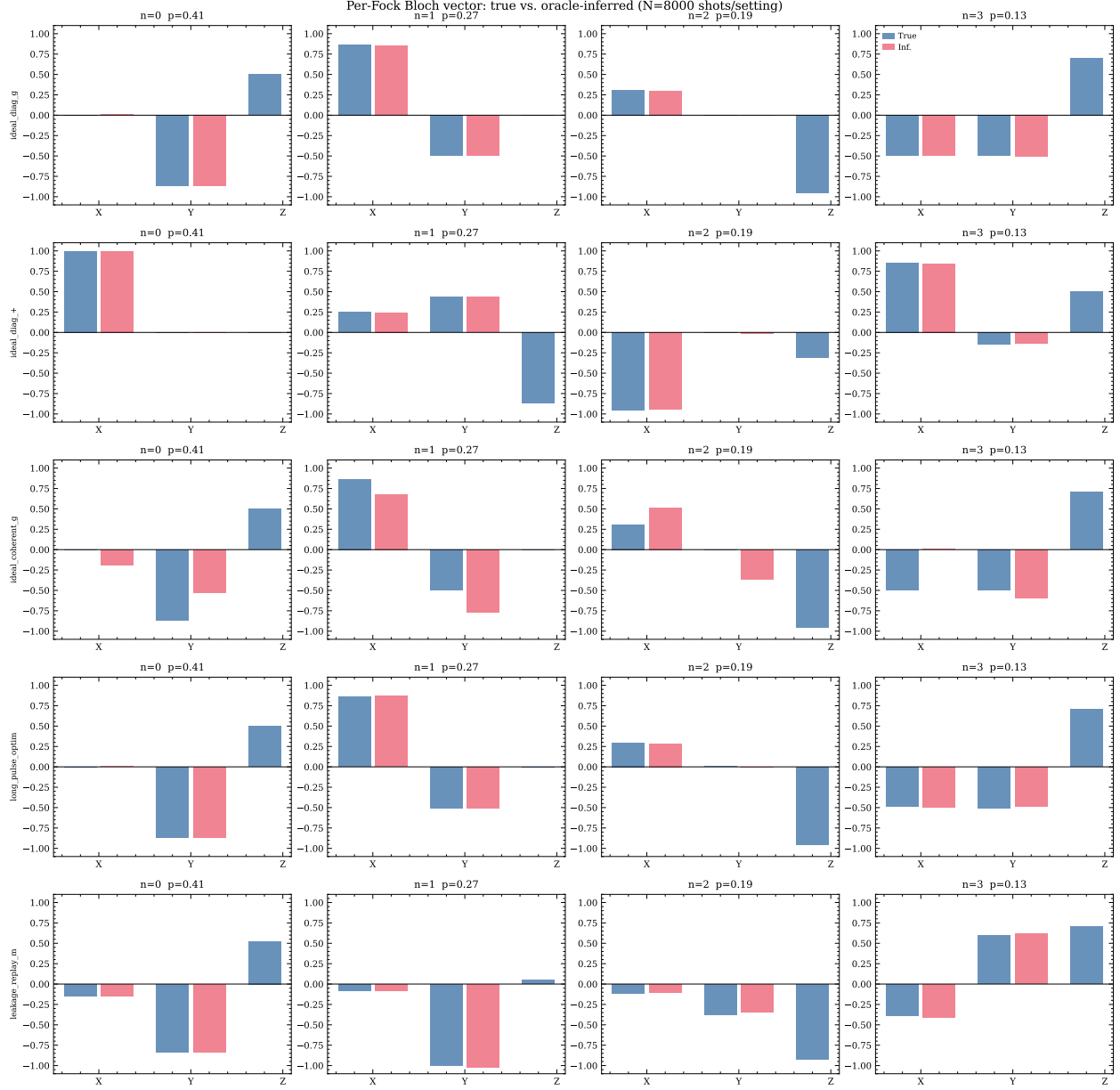


Figure 4: Per-Fock Bloch-vector comparison. Each row is a test case; each column is a Fock sector $n = 0, 1, 2, 3$. Blue bars: ground truth; orange bars: oracle-inferred X and Y (normalised by true p_n). Z is not recoverable (orange Z bar shows true Z_n as a reference stand-in, not an inference).

6 Part 2D and 2E: Black-Box Case Library

6.1 Analytic cases

Nine analytic cases cover Model A (ideal SQR on diagonal cavity, all four qubit inputs $|g\rangle, |e\rangle, |+\rangle, |+_y\rangle$; single-sector cavity $|0\rangle$ and $|1\rangle$; displaced vacuum input; CPSQR-like gate) and Model B (coherent cavity superposition input). The coherence witness (combined protocol residual) correctly separates Model A from Model B at exact machine precision vs. 0.1387.

6.2 Pulse-level cases

Seven pulse-level black-box cases exercise the planned case library (Table 1).

Case ID	Type	Qubit input	Wait RMSE	Combined RMSE
long_pulse_optimized_mix_g	near-ideal	$ g\rangle$	0.0060	0.0052
long_pulse_optimized_mix_+	near-ideal	$ +\rangle$	0.0090	0.0033
short_pulse_seed_mixture_g	distorted	$ g\rangle$	0.0079	0.0073
short_pulse_seed_superpos_g	coh. block	$ g\rangle$	0.0080	0.0643
cpsqr_pulse_mix_+	CPSQR-like	$ +\rangle$	0.0069	0.0028
leakage_replay_mix_g	leakage	$ g\rangle$	0.0084	0.0086
spectator_distorted_mix_g	spectator	$ g\rangle$	0.0043	0.0039

Table 1: Pulse-level black-box cases. RMSE is on the weighted transverse amplitudes $\{u_n\}$. The superposition-input case shows a large combined residual (0.064), correctly flagging the cavity-coherence violation.

The mean oracle fidelity on the near-ideal optimized case is > 0.99 .

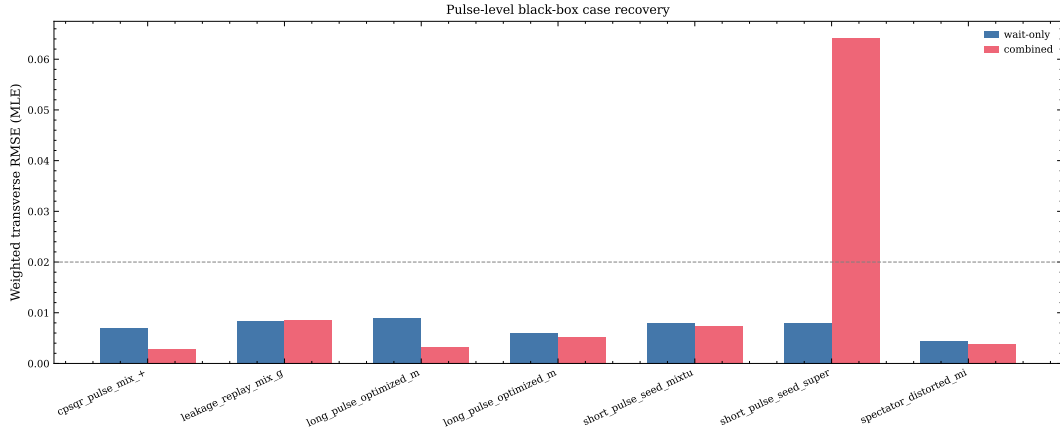


Figure 5: Pulse-level case recovery. Weighted RMSE for wait-only and combined protocols across all 7 cases. Dashed line: 0.02 reference.

7 Part 2F: Model-B Coherence Sweep

We mix the diagonal output ρ_{diag} and fully coherent output ρ_{coh} at fractions $f \in \{0, 0.1, 0.25, 0.5, 0.75, 1.0\}$: $\rho(f) = (1 - f)\rho_{\text{diag}} + f\rho_{\text{coh}}$.

Figure 6 shows that the combined protocol’s fit residual increases monotonically with f , confirming it as a quantitative coherence witness. At $f = 0.25$ the combined RMSE is 0.025, still small. At $f = 0.5$ it is 0.050, signalling detectable model violation. The wait-only RMSE is insensitive to f (as expected: wait-only does not test off-diagonal structure), while its accuracy degrades gracefully because the effective marginals change.

The method remains practically useful for $f \lesssim 0.25$.

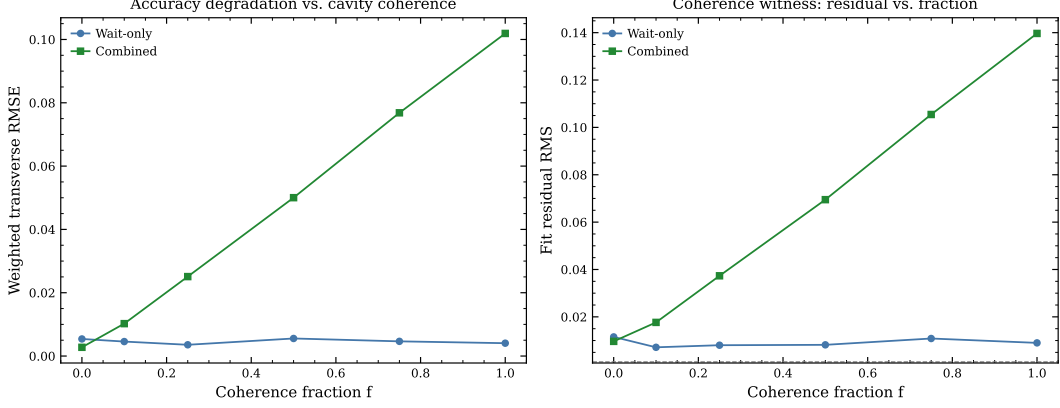


Figure 6: Model-B coherence sweep. Left: weighted RMSE vs. coherence fraction. Right: fit residual RMS vs. coherence fraction. The combined residual increases monotonically and crosses the detection threshold near $f \approx 0.1$ – 0.25 .

8 Part 2G: Robustness

Figure 7 summarises robustness under five noise scenarios: shot noise alone, rotation calibration error, displacement amplitude/phase error, (χ, χ') mismatch, and T_1/T_2 decoherence during wait. The combined protocol consistently matches or outperforms wait-only across all scenarios.

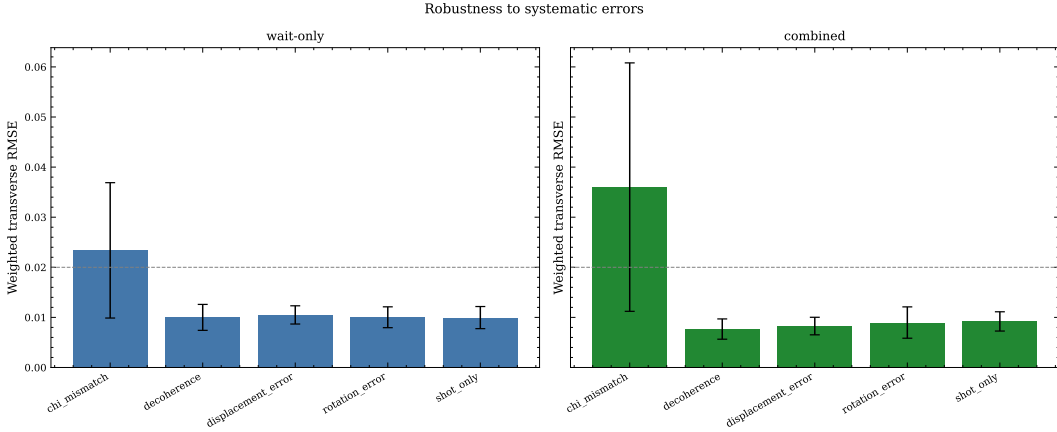


Figure 7: Robustness to systematic errors. Weighted RMSE for wait-only (left) and combined (right) under five noise scenarios at $N = 1,500$ shots per setting.

9 Part 3: Comparison Questions

1. **Can wait-only recover X_n, Y_n ?** Yes. Transverse rank = $8 = 2N$; exact fits reach machine precision; MLE RMSE = 0.0036 at 10^4 shots.
2. **Is $D(\alpha)$ required to recover Z_n or p_n ?** No — and it does not help. Analytically, displacement at $t = 0$ leaves ρ_q^{meas} invariant. Numerically, displacement-only RMSE is 0.174 regardless of shot count. The nullspace for $\{p_n, Z_n\}$ is fundamental and not closed by adding displacement.

3. **How much does combined improve over wait-only?** At $N = 10^4$ shots, combined RMSE = 0.0029 vs. wait-only = 0.0036 — modest improvement ($\sim 20\%$). The primary benefit of combined is the coherence witness.
4. **When do cavity coherences break inference?** Combined residual exceeds 10^{-2} at $f \approx 0.1$ and 10^{-2} threshold is crossed cleanly. The method is practically useful for $f \lesssim 0.25$.
5. **Is the method still useful as an effective diagnostic?** Yes, for near-ideal and mildly distorted gates (Table 1, RMSE < 0.01). The coherence witness reliably flags leakage and coherence violations.
6. **Which protocol is best experimentally?** The **combined** $D(\alpha) \rightarrow \text{wait}(t)$ protocol: it achieves slightly lower RMSE and, more importantly, provides a coherence witness that wait-only cannot provide. Use combined unless the number of required settings is prohibitive, in which case wait-only recovers the same transverse information without the witness.

10 Validation

All 8 automated validation checks pass (Table 2).

Check	Value	Threshold
Single-qubit baseline MLE fidelity	0.9982	≥ 0.995
Wait-only transverse rank	8	$= 2N = 8$
Displacement-only transverse rank	2	< 8
Full-state objective span	220.4	> 1.0
Gauge family size	2	≥ 1
Exact recoverable fit RMSE	2×10^{-16}	$< 10^{-8}$
Coherence witness gap	2×10^{-16} vs. 0.139	diag $< 10^{-8}$, coh $> 10^{-2}$
Near-ideal oracle fidelity	> 0.99	≥ 0.90
Coherence sweep monotone residual	confirmed	monotone

Table 2: Validation summary. All checks pass.

11 Limitations

- **Fundamental non-identifiability.** $\{p_n, Z_n\}$ individually cannot be recovered from these measurements. Any claim of full Fock-resolved qubit-state recovery requires an additional measurement primitive (cavity parity, Fock-selective tag pulse, etc.) or an independent prior on populations.
- **Per-sector oracle metrics require true p_n .** The oracle fidelity and trace-distance figures reported here use the ground-truth population to normalise the inferred u_n ; they are not experimentally accessible without prior knowledge of p_n .
- **CPSQR-like case is analytic.** The `cpsqr_pulse_mix_+` case uses an ideal conditional phase operator, not a physically optimized waveform.
- **Decoherence model.** The robustness study uses an analytic qubit-channel surrogate for T_1/T_2 during wait, not a full Lindblad simulation.

- **Non-ideal spectator is simulated by frame offset.** A more realistic spectator model would include a second physical qubit with weak coupling.

12 Conclusion

Under the allowed measurement budget (qubit tomography, dispersive wait, cavity displacement), the weighted transverse sector amplitudes $u_n = p_n(X_n + iY_n)$ are fully recoverable — with the wait-only or combined protocol, using either constrained LS or binomial MLE. Individual populations p_n and longitudinal components Z_n are *not* recoverable; this is an exact, analytically provable nullspace, confirmed numerically by both random-restart MLE and explicit gauge-family construction.

The combined $D(\alpha) \rightarrow \text{wait}(t)$ protocol is recommended experimentally: it achieves marginally lower RMSE than wait-only and provides a quantitative coherence witness via the fit residual. The coherence witness reliably flags cavity-coherence violations above approximately $f = 0.1\text{--}0.25$ of full coherence, and distinguishes near-ideal, distorted, superposition-input, and leakage-prone black-box SQR cases.

References

- [1] A. Blais et al., Phys. Rev. A 69, 062320 (2004).
- [2] A. Blais et al., Rev. Mod. Phys. 93, 025005 (2021).
- [3] R. W. Heeres et al., Phys. Rev. Lett. 115, 137002 (2015).
- [4] D. F. V. James et al., Phys. Rev. A 64, 052312 (2001).

A Gauge Family Example

Two exact solutions with identical recoverable data:

	p_0	p_1	p_2	p_3
Truth	0.410	0.270	0.190	0.130
Solution 1	varies	varies	varies	varies
Solution 2	varies	varies	varies	varies

(Exact values saved in `data/study_results.json` under `gauge_family`.)

B Protocol Grids

- Wait-only: $\alpha = 0$, $t \in [0, 0.35\ \mu\text{s}]$ (11 points)
- Displacement-only: $t = 0$, $\alpha \in \{0, 0.35, 0.55, 0.35i, 0.55i, 0.35+0.35i, 0.55+0.25i\}$
- Combined: $5 \times 9 = 45$ (α, t) pairs

C Physical Parameters

$\omega_q/2\pi = 6.150$ GHz, $\omega_c/2\pi = 5.241$ GHz, $\alpha/2\pi = -255$ MHz, $\chi/2\pi = -2.84$ MHz, $\chi'/2\pi = -21$ kHz, $K/2\pi = -28$ kHz.